POLS 4728 design language

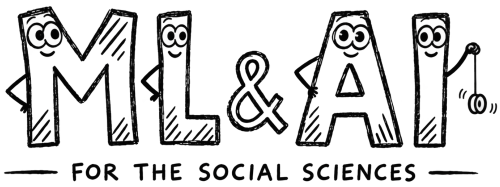
Scholarly notes with a playful visual identity.

A welcoming identity

Primary / outlined characters



Playful / the yo-yo variant



Use character artwork on covers and part openers, with occasional chapter illustrations that clarify a statistical idea. Develop concepts through explanation, mathematical reasoning, evidence, and code in connected paragraphs. On the notes cover, place the complete original yo-yo logo high on the page. Its embedded subtitle supplies the visible title; do not repeat it in a separate typeset heading. Retain the full course title in the PDF metadata.

Document accents and plotting colors

Blue and rust anchor the document. Plots use an eight-color cycle with matching colors for related marks, plus labels or patterns that survive grayscale printing.

Blue Rust Teal Purple

Ochre Sky Rose Gray

A familiar mathematical voice

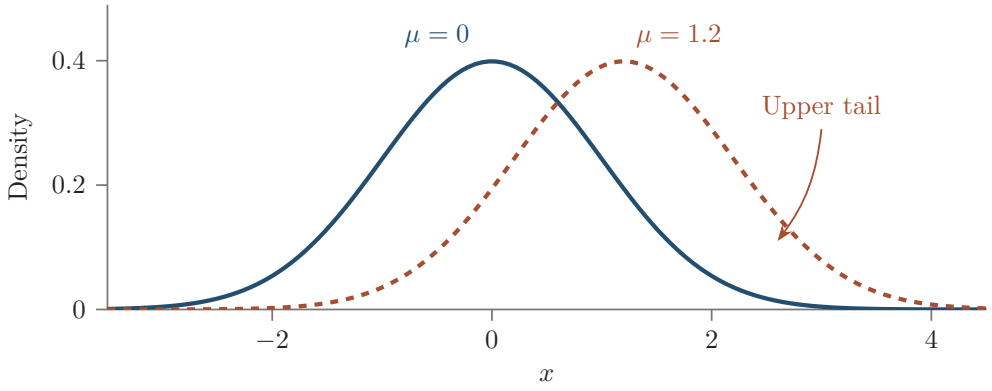
Latin Modern text and mathematics retain the character of a TeX textbook. Use clear hierarchy, open space, and readable labels at the final printed size.

$$\widehat{y}_i = f_\theta(x_i), \qquad \text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2.$$

Print defaults: white backgrounds, 11 pt body text, 10 pt plot labels, thin rules, and no decorative page textures.

Exposition, figures, and code

Open textbook, with a useful annotation



Two unit-variance normal distributions. Moving the mean from 0 to 1.2 shifts probability to the right. Labels and line patterns identify both curves even when the page is printed in grayscale.

ROC and precision–recall plots keep all four borders around $[0, 1]^2$, overriding the open-axis default. Label zooms as detail views.

Develop explanations in connected prose, preserving assumptions and derivations. Keep useful lists for vocabulary, assumptions, examples, comparisons, and procedures. Avoid fragmented paragraphs, forced line breaks, punchy lead-ins, decorative boxes, quirky pull quotes, story sidebars, and bold body emphasis. Use `\emph{}` for new vocabulary, such as *training data*, or necessary stress. Keep later mentions and routine lead-ins in regular type.

Code students can read and reuse

Use `listings` for selectable, monospaced text on white. Aim for at least 10 pt at printed size. Keep lines short, preserve indentation, and use line numbers only when the explanation needs them.

This example from the Python notes squares each prediction error and adds the results. The output is 14.

```
errors = [1, -2, 3]
total = 0
for error in errors:
    total += error ** 2
print(total)  # 14
```

Check readability in grayscale and paste a sample into an editor: indentation, punctuation, and line breaks must survive. Bold keywords and italic comments support the colors. Shorten long examples before reducing the type size.

Illustration as explanation

A reference from *Statistics*

Freedman, Pisani, and Purves's *Statistics* (4th ed., W. W. Norton, 2007) is a reference for the relationship between exposition and illustration: direct prose, concrete examples, economical diagrams, and occasional visual wit. Use sparse black line drawings on white backgrounds, with simple compositions and little shading. Within chapters, an illustration should clarify a mechanism, reveal a misconception, or make a comparison memorable. Humor should arise from the statistical idea.

Conceptual drawings and quantitative figures

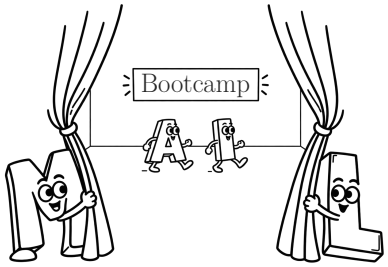
Hand-drawn character is welcome in conceptual illustrations; quantitative figures must retain precise geometry, readable labels, and faithful scales. Preserve the course's established typography, palette, and characters, and develop original illustrations suited to its subject matter. Conceptual drawings must not imply numerical relationships that the explanation does not establish.

A place within the argument

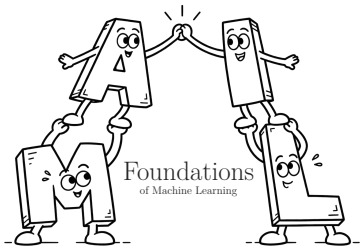
Place each illustration near the passage it explains. Its caption and the surrounding prose should state the learning point; essential labels and mathematical notation remain real text. Use illustrations selectively when they advance the argument. Keep the existing rules for scholarly prose, ordinary headings, and restrained formatting.

Use a clear silhouette and legible details at the final printed size. Register new character artwork in `design/assets.json` with a description, hash, and intended use, and keep image-generation prompts outside version control. Review the result in grayscale and describe meaningful visual content in alternative text where the format supports it.

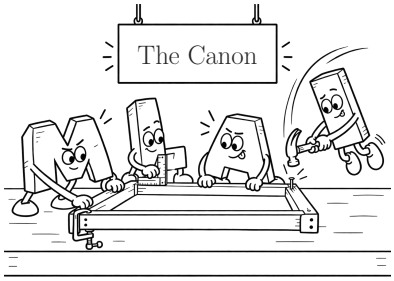
A welcoming start to each part.



M and L open the curtains; A and I march farther back on the stage.



M and L support A and I, whose joined hands complete an arch.



The letters build together, with I airborne during a hammer swing.

Typeset one title inside the unlettered scene using regular-weight Latin Modern. Keep it selectable and preserve the contents entry and PDF bookmark. Use a small part number above the composition and a quiet folio. Balance the whole group slightly above the page's center, at about 90% of the text width.

Preserve recognizable characters, sparse black outlines, and the full scene. Keep draft notices in introductory prose. Keep the original artwork and register derivatives in `design/assets.json`. All three openers use the same composition helper in the guide and notes.

Python source beside each listing

Use the *Matplotlib for Storytellers* pattern: a readable `.py` filename at the listing's upper right opens its GitHub source page. Five active listings use maintained local source files and 10 pt type. Their public copies live in `pols4728`, while the semester repository remains private.

The shared helper prints the local file and links to a matching public commit. When an example changes, publish its updated copy, update the shared revision, and rebuild the notes. Check that each link works without signing in and that the downloaded file matches the printed source.

Readable in more than one way

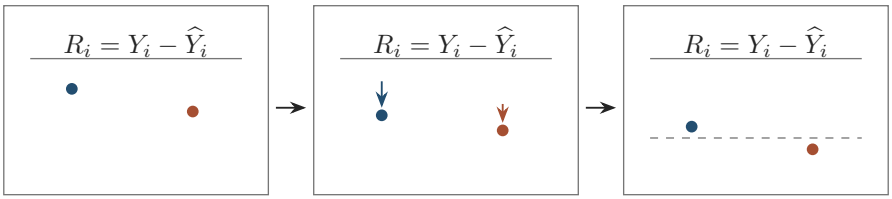
Pair color with words, shapes, and line patterns. Keep essential information in text and check pages at actual size and in grayscale.

Accessibility status. This system improves visual readability and authoring practice. The current PDFs are untagged; screen-reader structure, accessible mathematics, and embedded alternative text need a separate build migration. The complete rules and migration register are in `design/README.md`.

Scrollytelling and interactive explainers

Scrolling carries the explanation

Every course visual explainer must be a working scrollytelling experience unless the user explicitly requests another format. Pin a key diagram or equation while scroll position constructs or transforms it and reveals important new text. Storyboards and prototypes must demonstrate that motion. A long illustrated document, entrance fades alone, or timer-driven playback is insufficient.



A pinned equation anchors a changing diagram. These frames specify a conceptual progression, not a numerical dataset: the same marks move as scrolling subtracts predictions. The accompanying text explains each step.

Specify each scene

Record its learning point, pinned visual, scroll interval, transformation, revealed text, and completed state. Meaningful intermediate states must follow scroll position; backward scrolling revisits them. Preserve the identity of observations, scales, and color roles across related scenes. Hold each diagram long enough to read the text. Standalone lessons must not refer to the source assignment, quiz, or question number.

Keep the experience usable

Retain a compact pinned diagram or equation on narrow screens, with room for changing text. Provide keyboard navigation. With reduced motion, show complete, legible states at logical scroll thresholds without continuous movement; preserve the full argument and results.

Verify actual scrolling

Before delivery, scroll forward and backward in a browser on desktop and narrow mobile layouts. Check pinning, intermediate geometry, new text, reverse progress, keyboard access, reduced motion, labels, and final values. Final-state screenshots and unit tests alone are insufficient. See `design/README.md` for the full authoring and review requirements.